MIL-I-23053/14A
01 May 1986
SUPERSEDING
MIL-I-23053/14
20 May 1976

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, ETHYLENE-TETRAFLUOROETHYLENE FLUOROPOLYMER, SEMI-RIGID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

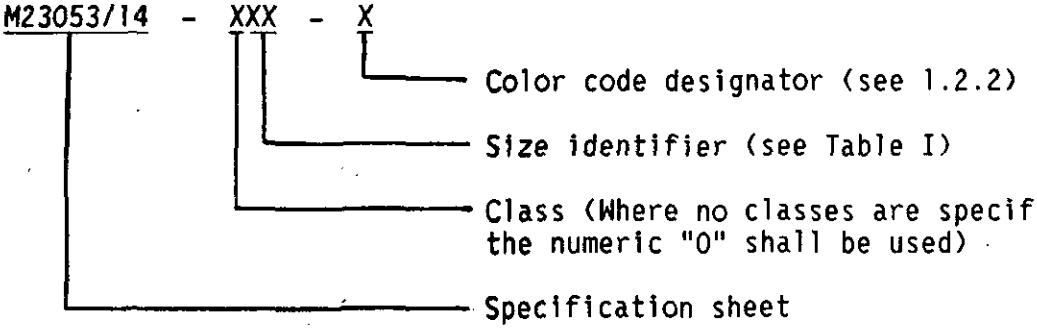
REQUIREMENTS:

Continuous operating temperature range: -100°C (-148°F) to 150°C (302°F).

Color: The sleeving shall be furnished in clear (translucent white to transparent blue), the as manufactured color.

Longitudinal change: ± 10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



Example: 0.500 inch (12.7 mm) as supplied ID ETFE sleeving shall be identified as M23053/14-006-C.

NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/14-001-C	.093 (2.4)	.062 (1.6)	.010 ± .002(.25 ± .05)
M23053/14-002-C	.125 (3.2)	.083 (3.1)	.010 ± .002(.25 ± .05)
M23053/14-003-C	.187 (4.7)	.125 (3.2)	.011 ± .002(.28 ± .05)
M23053/14-004-C	.250 (6.4)	.166 (4.2)	.013 ± .003(.33 ± .08)
M23053/14-005-C	.375 (9.5)	.250 (6.4)	.013 ± .003(.33 ± .08)
M23053/14-006-C	.500 (12.7)	.345 (8.3)	.013 ± .003(.33 ± .08)
M23053/14-007-C	.750 (19.0)	.500 (12.7)	.018 ± .004(.46 ± .10)
M23053/14-008-C	1.000 (25.4)	.665 (16.7)	.022 ± .004(.56 ± .10)
M23053/14-009-C	1.250 (31.8)	.835 (21.2)	.030 ± .004(.80 ± .10)
M23053/14-010-C	1.500 (38.0)	1.000 (25.4)	.030 ± .004(.80 ± .10)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

2/ Wall thickness dimensions are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5; 175° ± 1°C (347° ± 2°F) for 10 minutes.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID minimum	Table I	4.6.3
Restricted shrinkage	No cracks	4.6.6.12 175° ± 2°C (347° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability	Pass	4.6.17 175° ± 1°C (347° ± 2°F), 24 hours
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	6,500 (44.8)	4.6.14 ASTM D 638, 20 inches/minute
Ultimate elongation, percent, minimum	200	4.6.14 ASTM D 638, 20 inches/minute

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Dielectric breakdown, volts, minimum	11,000	4.6.2 ASTM D 876
Volume resistivity, Ohm-cm, minimum	10^{15}	4.6.2 ASTM D 876
Water absorption, percent, maximum	0.1	4.6.2 ASTM D 570, 24 hrs. at 23°C
Low temperature flexibility	No cracking	4.6.7.1 -65° ± 1°C (-85° ± 2°F)
Heat resistance, properties after:		4.6.9 200° ± 2°C (392° ± 4°F) for 168 hours
Tensile strength, psi (MPa), minimum	6,000 (41.4)	
Ultimate elongation, percent, minimum	175	
Dielectric breakdown volts, minimum	10,000	
Specific gravity, max	1.73	4.6.2 ASTM D 792
Flammability	Self-extinguishing 15 seconds and 3 inches	4.6.15.2

Shelf life conditions: Supplier shall certify to storage at 40°C (104°F) for four years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable, semi-rigid, ethylene-tetrafluoroethylene fluoropolymer is intended for use in component cables and electronic lead strain relief where low expansion ratios are satisfactory.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA, MI
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

(Project 5970-0577-13)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23053 /14A	INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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PREVIOUS EDITION IS OBSOLETE.

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